

High dimensional Probability

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1. SUB-GAUSSIAN RANDOM VARIABLES

2. CONCENTRATION INEQUALITIES

3. GAUSSIAN MATRICES

3.1. Gaussian Width

The Gaussian width can be seen as a measure of geometric complexity such as volume or surface area. Roughly, it tells you on average what the support function values will be if the normal directions are sample from an isotropic Gaussian distribution.

Definition 3.1.1 (Gaussian Width). Let $S \subset \mathbb{R}^n$ be a subset. The *Gaussian width* is defined as

$$w(S) = \mathbb{E} \left[\sup_{s \in S} \langle g, s \rangle \right] \text{ where } g \sim \mathcal{N}(0, 1)$$

3.2. Gordon's Comparison Theorem

Gordon's Comparison Theorem gives an similar analogue to the JL lemma in the sense that it describes the amount of distortion one should expect with high probability after randomly projecting a vector to a lower dimensional space:

Theorem 3.2.1 (Gordon's Comparison Theorem). Let $G \in \mathbb{R}^{k \times d}$ be a matrix whose entries are sampled from independent unit Gaussians $\mathcal{N}(0, 1)$ and $S \subset \mathbb{S}^{d-1}$ a closed subset of the unit sphere. For $t \geq 0$,

$$\lambda_m - w(S) - t \leq \inf_{s \in S} \|Gs\| \leq \sup_{s \in S} \|Gs\| \leq \lambda + w(S) + t$$

with probability at least $1 - 2 \exp\left(-\frac{t^2}{2}\right)$

Here, $w(S)$ refers to the Gaussian width of S . See Definition 3.1.1.

Proof.

□